

01 output:-

(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts\$ python 01_preprocess.py

[00:00:00] INFO

[00:00:00] INFO

[00:00:00] INFO §3.1 Loading and merging review CSVs

[00:00:00] INFO

[00:00:00] INFO Metadata loaded for 26 apps.

[00:00:00] INFO	Athan_Prayer_Times_Al_Quran	90,003 reviews
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[00:00:00] INFO	Deen_App	7,497 reviews
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[00:00:00] INFO	Everyday_Muslim_-_Pray_tracker	1,106 reviews
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[00:00:00] INFO	IslamApp_Prayer_times_Athan	577 reviews
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[00:00:00] INFO	Islamic_Habit_tracker	799 reviews
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[00:00:01] INFO	Muslim_Bangla_Quran_Hadith_Dua	53,332 reviews
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[00:00:01] INFO	Muslim_Muna	610 reviews
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[00:00:01] INFO	Muslim_Prayer_Qibla_Finder	2,541 reviews
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[00:00:01] INFO	Muslim_Pro	283,371 reviews
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[00:00:01] INFO	Muslims_Day-নমজ্জ_করআন_দয়	37,257 reviews
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[00:00:01] INFO	Niya	54 reviews
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[00:00:01] INFO	Pillars_Prayer_Times_Qibla	1,691 reviews
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[00:00:01] INFO	Pray_Watch	74 reviews
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[00:00:01] INFO	PrayerAlarm	5 reviews
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[00:00:01] INFO	Prayer_Time_Azan_Alarm_Qibla	8,392 reviews
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[00:00:01] INFO	Prayer_Times_-_Qibla_Namaz	65,085 reviews
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[00:00:01] INFO	Prayer_Times_-_Qibla_Salah	4,776 reviews
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[00:00:01] INFO	Prayer_Times_and_Qibla	4,546 reviews
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[00:00:01] INFO	Sadiq_Prayer_Quran_Qibla	1,525 reviews
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[00:00:01] INFO	Sajda_Quran_Athan_Prayer	5,135 reviews
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[00:00:01] INFO	Salaat_First_Prayer_Times	8,895 reviews
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[00:00:02] INFO	Salatuk_Prayer_time	111,232 reviews
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[00:00:02] INFO	WeMuslim	29,470 reviews
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[00:00:02] INFO	iPray_Prayer_Times_Qibla	3,355 reviews
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[00:00:02] INFO	iQIBLA_Life	4,294 reviews
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[00:00:02] INFO	নমজ্জর_সময়সচ_-_Prayer_Time	6,572 reviews
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[00:00:02] INFO Concatenated: 732,194 rows from 26 apps.

[00:00:03] INFO Merged corpus: 732,194 rows × 23 cols (833 MB)

[00:00:03] INFO

[00:00:03] INFO

[00:00:03] INFO §3.1 Text normalization

[00:00:03] INFO

[00:00:06] INFO Emoji present in 8.3% of reviews.
[00:00:06] INFO Empty after cleaning: 9,575
[00:00:06] INFO
[00:00:06] INFO

[00:00:06] INFO §3.1 Date parsing
[00:00:06] INFO

[00:00:06] INFO Temporal coverage: 2011-04 → 2026-08
[00:00:06] INFO Unparseable dates: 0
[00:00:06] INFO
[00:00:06] INFO

[00:00:06] INFO §3.1 Review-level features
[00:00:06] INFO

[00:00:07] INFO Median length: 4 words
[00:00:07] INFO Under 5 words: 55.7%
[00:00:07] INFO Developer reply rate: 17.3%
[00:00:07] INFO
[00:00:07] INFO

[00:00:07] INFO §3.2 Language detection
[00:00:07] INFO

[00:00:10] INFO Top languages:

[00:00:10] INFO	en	598,996	(81.8%)
[00:00:10] INFO	ar	38,107	(5.2%)
[00:00:10] INFO	bn	29,268	(4.0%)
[00:00:10] INFO	id	10,984	(1.5%)
[00:00:10] INFO	unk	9,575	(1.3%)
[00:00:10] INFO	ur	5,629	(0.8%)
[00:00:10] INFO	fr	5,475	(0.7%)
[00:00:10] INFO	de	3,165	(0.4%)
[00:00:10] INFO	it	3,026	(0.4%)
[00:00:10] INFO	ms	2,855	(0.4%)

[00:00:10] INFO Language mix per app (top 5 languages, share of that app's reviews):

[00:00:10] INFO	Athan: Prayer Times & Al Quran	en=92%	ar=2%	bn=0%
id=0%	unk=1%			
[00:00:10] INFO	Deen - Islamic App	en=87%	ar=0%	bn=8%
id=0%	unk=1%			
[00:00:10] INFO	Everyday Muslim - Pray tracker	en=93%	ar=1%	bn=0%
id=0%	unk=2%			
[00:00:10] INFO	IslamApp: Prayer times & Athan	en=88%	ar=1%	bn=0%

id=1% unk=3%				
[00:00:10] INFO	Islamic Habit Tracker	en=89%	ar=3%	bn=1%
id=1% unk=1%				
[00:00:10] INFO	Muslim Bangla Quran Hadith Dua	en=63%	ar=1%	
bn=30% id=0% unk=1%				
[00:00:10] INFO	Muslim Muna Adhan Prayer Quran	en=94%	ar=2%	bn=0%
id=0% unk=1%				
[00:00:10] INFO	Muslim Pro: Quran Athan Prayer	en=89%	ar=1%	bn=0%
id=3% unk=1%				
[00:00:10] INFO	Muslim: Prayer, Qibla Finder	en=83%	ar=2%	bn=3%
id=1% unk=2%				
[00:00:10] INFO	Muslims Day-Prayer, Quran, Dua	en=69%	ar=0%	
bn=25% id=0% unk=1%				
[00:00:10] INFO	Niya: Muslim prayer tracker	en=94%	ar=2%	bn=0%
id=2% unk=0%				
[00:00:10] INFO	Pillars: Prayer Times & Qibla	en=97%	ar=1%	bn=0%
id=0% unk=1%				
[00:00:10] INFO	Pray Watch	en=92%	ar=3%	bn=0%
id=0% unk=3%				
[00:00:10] INFO	Prayer Alarm	en=100%	ar=0%	
bn=0% id=0% unk=0%				
[00:00:10] INFO	Prayer Time, Azan Alarm, Qibla	en=88%	ar=1%	bn=1%
id=1% unk=2%				
[00:00:10] INFO	Prayer Times - Qibla & Namaz	en=85%	ar=0%	bn=0%
id=0% unk=1%				
[00:00:10] INFO	Prayer Times - Qibla & Salah	en=91%	ar=3%	bn=0%
id=1% unk=1%				
[00:00:10] INFO	Prayer Times and Qibla	en=78%	ar=7%	bn=0%
id=9% unk=1%				
[00:00:10] INFO	Sadiq: Prayer, Quran, Qibla	en=88%	ar=3%	bn=5%
id=1% unk=1%				
[00:00:10] INFO	Sajda: Prayer Times, Quran	en=78%	ar=3%	bn=0%
id=3% unk=3%				
[00:00:10] INFO	Salaat First: Prayer Times	en=54%	ar=35%	
bn=0% id=0% unk=5%				
[00:00:10] INFO	Wemu: WeMuslim, Athan&Qibla	en=80%	ar=9%	bn=1%
id=0% unk=3%				
[00:00:10] INFO	iMuslim: Prayer Times & Quran	en=71%	ar=0%	
bn=22% id=0% unk=1%				
[00:00:10] INFO	iPray: Prayer Times & Qibla	en=92%	ar=3%	bn=0%
id=1% unk=1%				
[00:00:10] INFO	iQIBLA Life	en=84%	ar=5%	bn=0%
id=4% unk=1%				
[00:00:10] INFO	صلاتك Salatuk (Prayer time)	en=70%	ar=23%	bn=0%
id=0% unk=1%				
[00:00:10] INFO				

[00:00:10] INFO

[00:00:10] INFO §3.2 Translation (sampled, for human-readable output only)

[00:00:10] INFO

[00:00:10] INFO Non-English reviews in the text corpus: 48,265

[00:00:10] INFO --translate-sample not set → skipping translation (02b scores these natively with twitter-xlm-roberta).

[00:00:10] INFO

[00:00:10] INFO

[00:00:10] INFO §3.3 Analysis sub-corpora

[00:00:10] INFO

[00:00:10] INFO Feature CSV encoding: 322 present, 302 explicit-absent ('x'), 0 blank.

[00:00:10] INFO Feature CSV: 26 apps – 26 annotated, 0 unannotated (features left as NaN).

[00:00:10] INFO Resolved to packages: 26/26

[00:00:11] INFO Apps matched to the annotation sheet: 26

[00:00:11] INFO

[00:00:11] INFO Sub-corpus sizes:

[00:00:11] INFO corpus_all 732,194 (100.0%)

[00:00:11] INFO corpus_text 324,687 (44.3%)

[00:00:11] INFO corpus_en 276,422 (37.8%)

[00:00:11] INFO corpus_trans 48,265 (6.6%)

[00:00:11] INFO corpus_annot 324,687 (44.3%)

[00:00:11] INFO

[00:00:11] INFO

[00:00:11] INFO Writing output

[00:00:11] INFO

[00:00:13] INFO master_reviews: 732,194 rows × 44 cols (1043 MB)

[00:00:13] INFO Saved → /home/ubuntu/Tonmoy/Salat-App-Research/src/data/master_reviews.parquet

[00:00:13] INFO

[00:00:13] INFO Next: python src/scripts/02a_sentiment_vader.py

02a output:-

(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts\$ python 02a_sentiment_vader.py

[00:00:00] INFO C_text subset: 324,687 reviews

[00:00:00] INFO

[00:00:00] INFO

```
[00:00:00] INFO  §4.1  VADER sentiment
[00:00:00] INFO
```

vader: 100%|

324687/324687 [00:18<00:00, 17140.11review/s]

```
[00:00:20] INFO  VADER label distribution (all C_text):
```

```
[00:00:20] INFO      Positive 67.7%
```

```
[00:00:20] INFO      Neutral  24.7%
```

```
[00:00:20] INFO      Negative  7.6%
```

```
[00:00:20] INFO  VADER label distribution (English only):
```

```
[00:00:20] INFO      Positive 78.2%
```

```
[00:00:20] INFO      Neutral  13.2%
```

```
[00:00:20] INFO      Negative  8.6%
```

```
[00:00:20] INFO    Saved 324,687 rows → /home/ubuntu/Tonmoy/Salat-App-Research/src/data/_checkpoints/vader/vader.parquet
```

[00:00:20] INFO

```
[00:00:20] INFO Next: python src/scripts/02b_sentiment_roberta.py
```

```
(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts$
```

02b output:-

```
(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts$ python 02b_sentiment_roberta.py
```

```
[00:00:01] INFO    Loaded master: 732,194 reviews
```

```
[00:00:01] INFO  C text subset: 324,687 reviews
```

```
[00:00:01] INFO Device: cuda - NVIDIA GeForce RTX 5090 (31.4 GB VRAM)
```

```
[00:00:01] INFO  Auto batch size for 'base' on 31.4 GB → 512
```

```
[00:00:01] INFO
```

```
[00:00:01] INFO
```

```
[00:00:01] INFO  §4.2 Transformer sentiment
```

```
[00:00:01] INFO
```

```
[00:00:01] INFO    en: 276,422 reviews → cardiffnlp/twitter-roberta-base-sentiment-latest
```

```
[00:00:02] INFO Loading cardiffnlp/twitter-roberta-base-sentiment-latest on
cuda...
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF TOKEN to enable higher rate limits and faster downloads.

```
[transformers] `torch dtype` is deprecated! Use `dtype` instead!
```

Loading weights: 100%|

| 201/201 [00:00<00:00, 9906.41it/s]

[transformers] RobertaForSequenceClassification LOAD REPORT from:
cardiffnlp/twitter-roberta-base-sentiment-latest

Key	Status	Details
roberta.pooler.dense.weight	UNEXPECTED	
roberta.pooler.dense.bias	UNEXPECTED	

Notes:

- UNEXPECTED: can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

[00:00:05] INFO labels: ['negative', 'neutral', 'positive']

en 1/6: 100%

| 50000/50000 [00:04<00:00, 11932.60item/s]

en 2/6: 100%

| 50000/50000 [00:04<00:00, 10388.80item/s]

en 3/6: 100%

| 50000/50000 [00:03<00:00, 14050.13item/s]

en 4/6: 100%

| 50000/50000 [00:03<00:00, 14015.71item/s]

en 5/6: 100%

| 50000/50000 [00:03<00:00, 12564.16item/s]

en 6/6: 100%

| 26422/26422 [00:02<00:00, 11272.45item/s]

[00:00:28] INFO multi: 48,265 reviews → cardiffnlp/twitter-xlm-roberta-
base-sentiment

[00:00:28] INFO Loading cardiffnlp/twitter-xlm-roberta-base-sentiment on
cuda...

Loading weights: 100%

| 201/201 [00:00<00:00, 5122.71it/s]

[00:00:31] INFO labels: ['negative', 'neutral', 'positive']

multi 1/1: 100%|

48265/48265 [00:04<00:00, 10348.39item/s]

[00:00:36] INFO Transformer label distribution:

[00:00:36] INFO Positive 72.3%

[00:00:36] INFO Neutral 11.9%

[00:00:36] INFO Negative 15.8%

[00:00:36] INFO

[00:00:36] INFO

[00:00:36] INFO Merging sentiment sources

[00:00:36] INFO

[00:00:38] INFO Merged VADER for 324,687 reviews.

[00:00:38] INFO

[00:00:38] INFO

[00:00:38] INFO §4.3 Star-rating proxy

[00:00:38] INFO

[00:00:38] INFO Star-proxy distribution:

[00:00:38] INFO Positive 90.7%

[00:00:38] INFO Neutral 2.6%

[00:00:38] INFO Negative 6.6%

[00:00:38] INFO Reviews with transformer sentiment: 324,687 / 732,194

[00:00:38] INFO

[00:00:38] INFO

[00:00:38] INFO Writing output

[00:00:38] INFO

[00:00:40] INFO master_reviews_with_sentiment: 732,194 rows × 57 cols (1188 MB)

[00:00:40] INFO Saved → /home/ubuntu/Tonmoy/Salat-App-Research/src/data/master_reviews_with_sentiment.parquet

[00:00:40] INFO

[00:00:40] INFO Next: notebooks/02c_cross_validation.py, then python src/scripts/03_absa.py

(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts\$

03_absa output:-

(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts\$ python3 03_absa.py

[00:00:00] INFO C_text subset: 324,687 reviews
[00:00:00] INFO Device: cuda – NVIDIA GeForce RTX 5090 (31.4 GB VRAM)
[00:00:00] INFO
[00:00:00] INFO

[00:00:00] INFO §5.1–5.2 Aspect lexicon
[00:00:00] INFO

[00:00:01] INFO Lexicon: 27 aspects, 370 keywords, 105 regex patterns.
[00:00:01] INFO
[00:00:01] INFO

[00:00:01] INFO §5.3.1 Sentence segmentation
[00:00:01] INFO

[00:00:01] INFO Using pysbd segmenter.
segment: 100%|



324687/324687 [01:06<00:00, 4910.19review/s]

[00:01:07] INFO 324,687 reviews → 558,178 sentences (1.7 per review)
[00:01:07] INFO
[00:01:07] INFO

[00:01:07] INFO §5.3.2 Lexicon tagging
[00:01:07] INFO

[00:01:11] INFO	prayer_times_accuracy	23,482 sentences (4.21%)
[00:01:14] INFO	madhab	521 sentences (0.09%)
[00:01:17] INFO	calc_method	673 sentences (0.12%)
[00:01:20] INFO	reminders_adhan	33,502 sentences (6.00%)
[00:01:23] INFO	forbidden_times	1,628 sentences (0.29%)
[00:01:26] INFO	nafl_times	993 sentences (0.18%)
[00:01:29] INFO	goal_system	6,157 sentences (1.10%)
[00:01:31] INFO	mosque_finder	3,181 sentences (0.57%)
[00:01:34] INFO	guides	1,731 sentences (0.31%)
[00:01:37] INFO	adhkar	9,788 sentences (1.75%)
[00:01:40] INFO	qibla	8,906 sentences (1.60%)
[00:01:42] INFO	table_format	1,519 sentences (0.27%)
[00:01:46] INFO	qasr_travel	2,153 sentences (0.39%)
[00:01:48] INFO	calendar_sync	309 sentences (0.06%)
[00:01:52] INFO	monetization	16,549 sentences (2.96%)
[00:01:56] INFO	prayer_tracker	6,404 sentences (1.15%)
[00:02:00] INFO	tracker_score	934 sentences (0.17%)
[00:02:05] INFO	women_period	2,360 sentences (0.42%)
[00:02:07] INFO	widgets	5,260 sentences (0.94%)

```
[00:02:12] INFO      companion_hardware      2,711 sentences (0.49%)
[00:02:17] INFO      ui_design      16,625 sentences (2.98%)
[00:02:22] INFO      stability_bugs      8,104 sentences (1.45%)
[00:02:26] INFO      ads_intrusive      21,588 sentences (3.87%)
[00:02:31] INFO      privacy_data      8,078 sentences (1.45%)
[00:02:34] INFO      quran_audio      27,348 sentences (4.90%)
[00:02:39] INFO      complexity_bloat      5,707 sentences (1.02%)
[00:02:44] INFO      spiritual_affect      3,876 sentences (0.69%)
[00:02:44] INFO  Lexicon hits: 220,087 (aspect, sentence) pairs covering
168,821 / 558,178 sentences (30.2%)
[00:02:44] INFO  Auto batch size for 'large' on 31.4 GB → 512
[00:02:44] INFO
[00:02:44] INFO
```

```
[00:02:44] INFO  §5.3.3 Zero-shot backstop
[00:02:44] INFO
```

```
[00:02:45] INFO  Unmatched sentences eligible for zero-shot: 319,502
[00:02:46] INFO  Loading NLI model facebook/bart-large-mnli on cuda...
Warning: You are sending unauthenticated requests to the HF Hub. Please set
a HF_TOKEN to enable higher rate limits and faster downloads.
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|
```

```
| 515/515 [00:00<00:00, 5898.73it/s]
```

```
[00:02:48] INFO      entailment logit index: 2 ({0: 'contradiction', 1:
'neutral', 2: 'entailment'})
```

```
[00:02:48] INFO  Loading prefilter encoder sentence-transformers/all-MiniLM-
L6-v2 (top_k=5)...
```

```
Loading weights: 100%|
```

```
| 103/103 [00:00<00:00, 14354.51it/s]
```

```
[00:02:54] INFO  Zero-shot: 319,502 sentences → 289,051 unique (10%
collapsed)
```

```
Batches: 100%|
```

```
| 565/565 [00:16<00:00, 34.02it/s]
```

```
[00:03:11] INFO  Zero-shot: 1,445,255 NLI pairs (vs 7,804,377 without
prefilter)
```

```
nli: 100%|
```

```
| 1445255/1445255 [10:13<00:00, 2354.17item/s]
```

[00:13:27] INFO Zero-shot added 75,038 (aspect, sentence) pairs at threshold 0.75 across 56,146 reviews.

[00:13:27] INFO Zero-shot aspect distribution:

[00:13:27] INFO	quran_audio	26,975
[00:13:27] INFO	reminders_adhan	18,899
[00:13:27] INFO	spiritual_affect	7,523
[00:13:27] INFO	adhkar	6,366
[00:13:27] INFO	companion_hardware	3,328
[00:13:27] INFO	prayer_times_accuracy	2,955
[00:13:27] INFO	stability_bugs	1,658
[00:13:27] INFO	mosque_finder	1,013
[00:13:27] INFO	privacy_data	949
[00:13:27] INFO	complexity_bloat	878
[00:13:27] INFO		
[00:13:27] INFO		

[00:13:27] INFO §5.3.5 Request and negation handling

[00:13:27] INFO

/home/ubuntu/Tonmoy/Salat-App-Research/src/scripts/03_absa.py:282:

UserWarning: This pattern is interpreted as a regular expression, and has match groups. To actually get the groups, use str.extract.

```
tagged_df["is_request"] = text.str.contains(REQUEST_RE, regex=True, na=False)
```

/home/ubuntu/Tonmoy/Salat-App-Research/src/scripts/03_absa.py:283:

UserWarning: This pattern is interpreted as a regular expression, and has match groups. To actually get the groups, use str.extract.

```
tagged_df["has_negation"] = text.str.contains(NEGATION_RE, regex=True, na=False)
```

[00:13:29] INFO Flagged as requests: 17,758 (6.0%) – excluded from aspect sentiment, routed to 03c.

[00:13:29] INFO Containing negation: 53,601 (18.2%)

[00:13:29] INFO Auto batch size for 'base' on 31.4 GB → 512

[00:13:29] INFO

[00:13:29] INFO

[00:13:29] INFO §5.3.4 Per-aspect sentiment

[00:13:29] INFO

[00:13:29] INFO Scoring 277,367 non-request (aspect, sentence) pairs.

[00:13:29] INFO Unique sentences to score: 208,746 (25% collapsed)

[00:13:29] INFO Loading cardiffnlp/twitter-xlm-roberta-base-sentiment on cuda...

Loading weights: 100%|

```
| 201/201 [00:00<00:00, 5535.42it/s]
[00:13:32] INFO      labels: ['negative', 'neutral', 'positive']
aspect-sent 1/2: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
200000/200000 [00:11<00:00, 17373.48item/s]
aspect-sent 2/2: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
8746/8746 [00:00<00:00, 18597.95item/s]
[00:13:51] INFO    Aggregated to 254,802 (reviewId, aspect) rows.
[00:13:51] INFO    Aspect sentiment distribution:
[00:13:51] INFO          Positive  54.7%
[00:13:51] INFO          Neutral   16.8%
[00:13:51] INFO          Negative  28.5%
/home/ubuntu/Tonmoy/Salat-App-Research/src/scripts/03_absa.py:468:
FutureWarning: The behavior of DataFrame concatenation with empty or all-NA
entries is deprecated. In a future version, this will no longer exclude
empty or all-NA columns when determining the result dtypes. To retain the
old behavior, exclude the relevant entries before the concat operation.
    out = pd.concat([scored, _request_rows(tagged)], ignore_index=True)
[00:13:51] INFO
[00:13:51] INFO

[00:13:51] INFO    Writing output
[00:13:51] INFO

[00:13:52] INFO    aspect_sentiments: 272,251 rows × 15 cols (165 MB)
[00:13:52] INFO    Saved → /home/ubuntu/Tonmoy/Salat-App-
Research/src/data/aspect_sentiments.parquet
[00:13:52] INFO
[00:13:52] INFO    Reviews with >=1 aspect: 178,767
[00:13:52] INFO    Request-flagged rows:      17,449
[00:13:52] INFO
[00:13:52] INFO    Next: python src/scripts/03b_promise_extraction.py
(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-
Research/src/scripts$
```

```
(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-
Research/src/scripts$ python 03b_promise_extraction.py --no-zero-shot
[00:00:00] INFO
[00:00:00] INFO

[00:00:00] INFO    §6.1 Loading app descriptions
[00:00:00] INFO
```

```

[00:00:00] INFO  26 apps; description length median 2602 chars (range 951-
4137)
[00:00:00] INFO  Lexicon: 27 aspects, 370 keywords, 105 regex patterns.
[00:00:00] INFO  Promise variant: 7 aspects use a stricter vocabulary
(adhkar, companion_hardware, goal_system, guides, prayer_tracker,
qasr_travel, women_period)
[00:00:00] INFO
[00:00:00] INFO

```

```

[00:00:00] INFO  §6.1 Classifying description segments
[00:00:00] INFO

```

```

[00:00:00] INFO  Features claimed per app (description-derived):
[00:00:00] INFO      Sadiq: Prayer, Quran, Qibla                      13/20
[00:00:00] INFO      iPray: Prayer Times & Qibla                    12/20
[00:00:00] INFO      Athan: Prayer Times & Al Quran                  11/20
[00:00:00] INFO      iMuslim: Prayer Times & Quran                  11/20
[00:00:00] INFO      Deen - Islamic App                             11/20
[00:00:00] INFO      Pillars: Prayer Times & Qibla                  9/20
[00:00:00] INFO      Everyday Muslim - Pray tracker                 8/20
[00:00:00] INFO      IslamApp: Prayer times & Athan                 8/20
[00:00:00] INFO      Islamic Habit Tracker                        8/20
[00:00:00] INFO      Prayer Times and Qibla                        8/20
[00:00:00] INFO      Muslim Muna Adhan Prayer Quran                 7/20
[00:00:00] INFO      Salaat First: Prayer Times                    7/20
[00:00:00] INFO      Muslim: Prayer, Qibla Finder                  7/20
[00:00:00] INFO      Muslims Day-Prayer, Quran, Dua                7/20
[00:00:00] INFO      Prayer Time, Azan Alarm, Qibla                7/20
[00:00:00] INFO      Prayer Times - Qibla & Salah                  7/20
[00:00:00] INFO      Pray Watch                                    6/20
[00:00:00] INFO      Muslim Pro: Quran Athan Prayer                 6/20
[00:00:00] INFO      Niya: Muslim prayer tracker                  6/20
[00:00:00] INFO      Prayer Times - Qibla & Namaz                  6/20
[00:00:00] INFO      Muslim Bangla Quran Hadith Dua                5/20
[00:00:00] INFO      Wemu: WeMuslim, Athan&Qibla                  5/20
[00:00:00] INFO      Sajda: Prayer Times, Quran                    5/20
[00:00:00] INFO      iQIBLA Life                                    4/20
[00:00:00] INFO      صَلَاتُك Salatuk (Prayer time)                  4/20
[00:00:00] INFO      Prayer Alarm                                    3/20
[00:00:00] INFO  Feature CSV encoding: 322 present, 302 explicit-absent
('x'), 0 blank.
[00:00:00] INFO  Feature CSV: 26 apps – 26 annotated, 0 unannotated
(features left as NaN).
[00:00:00] INFO  Resolved to packages: 26/26
[00:00:00] INFO

```

[00:00:00] INFO

[00:00:00] INFO §6.1 Validation against hand annotation (Cohen's κ)

[00:00:00] INFO

[00:00:00] INFO Comparing against 26 hand-annotated apps.

[00:00:01] INFO	feature	n	csv	desc	agree
kappa					
[00:00:01] INFO 0.62	Women tracking	26	3	3	92%
[00:00:01] INFO 0.59	Forbidden Times	26	11	8	81%
[00:00:01] INFO 0.57	Salah and/or Wudu Guides	26	7	5	85%
[00:00:01] INFO 0.47	Qibla Compass	26	25	23	92%
[00:00:01] INFO 0.46	Prayer times in Table format	26	16	13	73%
[00:00:01] INFO 0.43	Goal System and Other events (st	26	7	4	81%
[00:00:01] INFO 0.39	Tracker Analysis / Score	26	9	8	73%
[00:00:01] INFO 0.39	Mosque Finder	26	12	14	69%
[00:00:01] INFO 0.36	Has Companion Hardware	26	1	4	88%
[00:00:01] INFO 0.34	Prayer Tracker	26	11	7	69%
[00:00:01] INFO 0.31	Salah Specific Adhkars	26	12	13	65%
[00:00:01] INFO 0.24	Widgets	26	19	10	58%
[00:00:01] INFO 0.19	Nafl Prayer Times	26	11	9	62%
[00:00:01] INFO 0.10	All Features for free	26	24	10	46%
[00:00:01] INFO 0.08	Various methods of calculation	26	24	9	42%
[00:00:01] INFO 0.01	Madhab Variations	26	25	3	15%
[00:00:01] INFO 0.00	Prayer Times by Location	26	26	22	85%
[00:00:01] INFO 0.00	Auto Qasr mode	26	1	0	96%
[00:00:01] INFO 0.00	Timely Reminders	26	26	25	96%

```

[00:00:01] INFO      Connect to Google Calendar      26      2      1      88%
-0.05
[00:00:01] INFO
[00:00:01] INFO  Mean Cohen's k across features: 0.275
[00:00:01] INFO  Interpretation: <0.20 poor · 0.21–0.40 fair · 0.41–0.60
moderate · 0.61–0.80 substantial · >0.80 near-perfect
[00:00:01] INFO  Saved k report → /home/ubuntu/Tonmoy/Salat-App-
Research/src/data/promise_source_agreement.csv
[00:00:01] INFO
[00:00:01] INFO

```

```

[00:00:01] INFO  §6.1 Monetization claim check
[00:00:01] INFO

```

```

[00:00:01] INFO  Apps claiming free features while carrying ads or IAP: 5
[00:00:01] INFO      Athan: Prayer Times & Al Quran      ads=True
iap=True
[00:00:01] INFO      Muslim Pro: Quran Athan Prayer      ads=True
iap=True
[00:00:01] INFO      Muslims Day-Prayer, Quran, Dua      ads=True
iap=True
[00:00:01] INFO      Prayer Time, Azan Alarm, Qibla      ads=True
iap=True
[00:00:01] INFO      Sajda: Prayer Times, Quran      ads=False
iap=True
[00:00:01] INFO
[00:00:01] INFO

```

```

[00:00:01] INFO  Combining sources
[00:00:01] INFO

```

```

[00:00:01] INFO  feature_count per app (CSV primary, description fallback):
[00:00:01] INFO      Deen - Islamic App      14/20
[csv]
[00:00:01] INFO      Muslims Day-Prayer, Quran, Dua      14/20
[csv]
[00:00:01] INFO      Athan: Prayer Times & Al Quran      13/20
[csv]
[00:00:01] INFO      Everyday Muslim - Pray tracker      13/20
[csv]
[00:00:01] INFO      Muslim Bangla Quran Hadith Dua      13/20
[csv]
[00:00:01] INFO      Islamic Habit Tracker      12/20
[csv]
[00:00:01] INFO      Muslim Pro: Quran Athan Prayer      12/20
[csv]

```

```

[00:00:01] INFO      Prayer Time, Azan Alarm, Qibla      12/20
[csv]
[00:00:01] INFO      Prayer Times - Qibla & Namaz      11/20
[csv]
[00:00:01] INFO      Wemu: WeMuslim, Athan&Qibla      11/20
[csv]
[00:00:01] INFO      Sadiq: Prayer, Quran, Qibla      11/20
[csv]
[00:00:01] INFO      iMuslim: Prayer Times & Quran      10/20
[csv]
[00:00:01] INFO      Muslim Muna Adhan Prayer Quran      10/20
[csv]
[00:00:01] INFO      Sajda: Prayer Times, Quran      10/20
[csv]
[00:00:01] INFO      Salaat First: Prayer Times      10/20
[csv]
[00:00:01] INFO      Pray Watch      10/20
[csv]
[00:00:01] INFO      Pillars: Prayer Times & Qibla      10/20
[csv]
[00:00:01] INFO      IslamApp: Prayer times & Athan      9/20
[csv]
[00:00:01] INFO      Muslim: Prayer, Qibla Finder      9/20
[csv]
[00:00:01] INFO      iQIBLA Life      9/20
[csv]
[00:00:01] INFO      Prayer Times and Qibla      9/20
[csv]
[00:00:01] INFO      Prayer Times - Qibla & Salah      9/20
[csv]
[00:00:01] INFO      صلاة Salatuk (Prayer time)      9/20
[csv]
[00:00:01] INFO      Niya: Muslim prayer tracker      8/20
[csv]
[00:00:01] INFO      iPray: Prayer Times & Qibla      8/20
[csv]
[00:00:01] INFO      Prayer Alarm      6/20
[csv]
[00:00:01] INFO      Saved → /home/ubuntu/Tonmoy/Salat-App-
Research/src/data/feature_matrix_combined.parquet
[00:00:01] INFO      Mean κ (CSV vs description): 0.275
[00:00:01] INFO
[00:00:01] INFO      Next: python src/scripts/03c_demand_mining.py
(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-
Research/src/scripts$

```

03c output:-

(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-Research/src/scripts\$ python 03c_demand_mining.py

[00:00:00] INFO C_text subset: 324,687 reviews

[00:00:00] INFO Lexicon: 27 aspects, 370 keywords, 105 regex patterns.

[00:00:00] INFO

[00:00:00] INFO

[00:00:00] INFO Sentence explosion

[00:00:00] INFO

[00:00:01] INFO 324,687 reviews → 501,335 candidate sentences

[00:00:01] INFO

[00:00:01] INFO

[00:00:01] INFO §6.2 Pattern extraction

[00:00:01] INFO

[00:00:02] INFO request 15,280 sentences (14,507 reviews)

[00:00:03] INFO absence 3,243 sentences (3,186 reviews)

[00:00:04] INFO churn 4,357 sentences (4,251 reviews)

[00:00:04] INFO preference 4,265 sentences (4,261 reviews)

[00:00:05] INFO tenure 725 sentences (720 reviews)

[00:00:05] INFO

[00:00:05] INFO

[00:00:05] INFO Attaching aspects

[00:00:05] INFO

[00:00:12] INFO Aspect-attached: 19,356 rows; unattached: 13,807 rows

[00:00:12] INFO Top aspects by demand volume:

[00:00:12] INFO reminders_adhan 3,248

[00:00:12] INFO quran_audio 3,092

[00:00:12] INFO prayer_times_accuracy 1,761

[00:00:12] INFO ads_intrusive 1,642

[00:00:12] INFO monetization 1,385

[00:00:12] INFO adhkar 1,109

[00:00:12] INFO widgets 883

[00:00:12] INFO ui_design 765

[00:00:12] INFO privacy_data 705

[00:00:12] INFO qibla 617

[00:00:12] INFO prayer_tracker 597

[00:00:12] INFO stability_bugs 476

[00:00:12] INFO mosque_finder 432

[00:00:12] INFO complexity_bloat 384

[00:00:12] INFO nafl_times 321

```
[00:00:12] INFO Agreement with ABSA is_request on request-type rows: 55.6%
[00:00:12] INFO
[00:00:12] INFO

[00:00:12] INFO Demand summary
[00:00:12] INFO

[00:00:12] INFO Total demand signals: 33,163 across 25,796 reviews
[00:00:12] INFO
[00:00:12] INFO By request type:
[00:00:12] INFO request 18,508
[00:00:12] INFO churn 5,018
[00:00:12] INFO preference 4,709
[00:00:12] INFO absence 4,117
[00:00:12] INFO tenure 811
[00:00:12] INFO
[00:00:12] INFO Rare-feature demand (the §2.1 rescue – these drive
RQ3/RQ4/RQ5):
[00:00:12] INFO qasr_travel 84 signals across 10 apps, 83
reviews
[00:00:12] INFO women_period 272 signals across 19 apps, 258
reviews
[00:00:12] INFO companion_hardware 314 signals across 19 apps, 298
reviews
[00:00:12] INFO mosque_finder 432 signals across 18 apps, 419
reviews
[00:00:12] INFO calendar_sync 94 signals across 15 apps, 93
reviews
[00:00:12] INFO
[00:00:12] INFO

[00:00:12] INFO §6.2 Precision-check sample
[00:00:12] INFO

[00:00:12] INFO Wrote 200 rows → /home/ubuntu/Tonmoy/Salat-App-
Research/src/data/gold_labels/demand_precision_sample.csv
[00:00:12] INFO ACTION: fill is_true_positive (1/0) and report precision
per request_type in the paper.
[00:00:12] INFO
[00:00:12] INFO

[00:00:12] INFO Writing output
[00:00:12] INFO

[00:00:12] INFO demand: 33,163 rows × 7 cols (13 MB)
[00:00:12] INFO Saved → /home/ubuntu/Tonmoy/Salat-App-
```

```
Research/src/data/demand.parquet
```

```
[00:00:12] INFO
```

```
[00:00:12] INFO Next: python src/scripts/04_topic_modeling.py
```

```
(Salat-App-Research) ubuntu@ubuntu-System-Product-Name:~/Tonmoy/Salat-App-  
Research/src/scripts$
```